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4.5 Homeostasis and Response

AQA GCSE Biology (8461) — Paper 2

4.5.1 Homeostasis

> **Homeostasis** is the regulation of the internal conditions of a cell or organism to maintain optimum conditions for function, in response to internal and external change. It maintains optimum conditions for **enzyme action** and all cell processes.

In humans, the main conditions controlled are:

- Body temperature (~37 °C)
- Blood glucose concentration
- Water & ion content (osmotic balance)

Components of a Control System

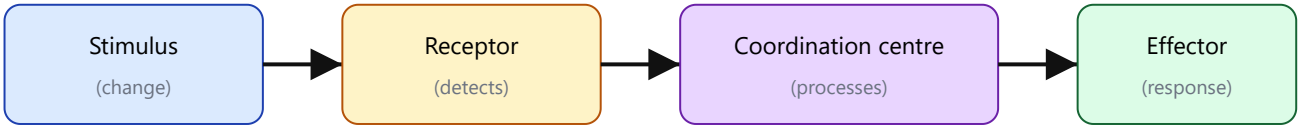

Fig 1: Components of a control system with negative feedback.

4.5.2 The Nervous System

The nervous system enables humans to react to their surroundings and coordinate behaviour. Nerve impulses travel along neurones, carrying electrical signals.

Reflex Arc

> A **reflex action** is a rapid, automatic response that does not involve conscious thought. Reflex arcs protect the body (e.g. withdrawing from a hot object, pupil reflex).

Stimulus → receptor → **sensory neurone** → **relay neurone** (in CNS/spinal cord) → **motor neurone** → effector (muscle/gland) → response.

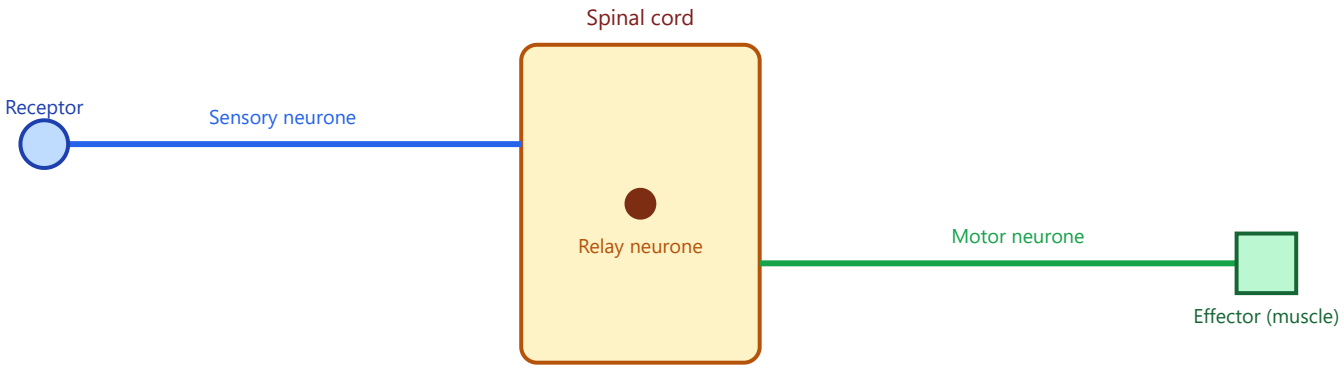

Fig 2: A reflex arc. Note that the impulse bypasses conscious thought — this is why it is fast.

Synapses

A synapse is the gap between two neurones. An electrical impulse triggers the release of **neurotransmitter chemicals**, which diffuse across the gap and trigger a new impulse in the next neurone.

4.5.3 The Brain (Separate Biology)

Region	Function
Cerebral cortex / cerebrum	Consciousness, memory, language, higher thought, voluntary movement

Cerebellum	Balance, coordination of movement
Medulla	Unconscious actions: heart rate, breathing rate
Hypothalamus	Regulates temperature, hormones, hunger
Pituitary	"Master gland" — secretes many hormones

4.5.4 The Eye (Separate Biology)

Key parts: cornea, iris, pupil, lens, ciliary muscles, suspensory ligaments, retina, optic nerve.

Accommodation

- **Distant object:** ciliary muscles relax, suspensory ligaments tight, lens thin — less refraction.
- **Near object:** ciliary muscles contract, suspensory ligaments slacken, lens thick — more refraction.

Eye Defects

- **Myopia (short sight)** — eyeball too long; image forms in front of retina. Corrected by *concave/diverging* lens.
- **Hyperopia (long sight)** — eyeball too short; image forms behind retina. Corrected by *convex/converging* lens.

4.5.5 Control of Body Temperature (Separate Biology)

The **hypothalamus** monitors blood temperature and signals responses.

Too hot	Too cold
Sweating (evaporation cools skin)	Shivering (muscle contraction releases heat)
Vasodilation (arterioles widen — more heat loss)	Vasoconstriction (arterioles narrow — less heat loss)

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Hairs lie flat	Hair erector muscles contract (trap air — insulation)

4.5.6 The Endocrine System and Hormones

Hormones are chemical messengers produced by glands, transported in the **blood** to target organs.

Gland	Main hormone(s)	Effect
Pituitary	FSH, LH, ADH, growth hormone, TSH	"Master gland" — controls other glands
Thyroid	Thyroxine	Raises metabolic rate
Pancreas	Insulin, glucagon	Controls blood glucose
Adrenal glands	Adrenaline	"Fight or flight" response
Ovaries	Oestrogen, progesterone	Menstrual cycle, secondary sexual characteristics (F)
Testes	Testosterone	Sperm production, secondary sexual characteristics (M)

4.5.7 Control of Blood Glucose

The **pancreas** detects blood glucose concentration and releases:
• **Insulin** when blood glucose is too *high* (after a meal) — causes cells to take up glucose and store as glycogen.
• **Glucagon** when blood glucose is too *low* — causes liver to break down glycogen back to glucose.

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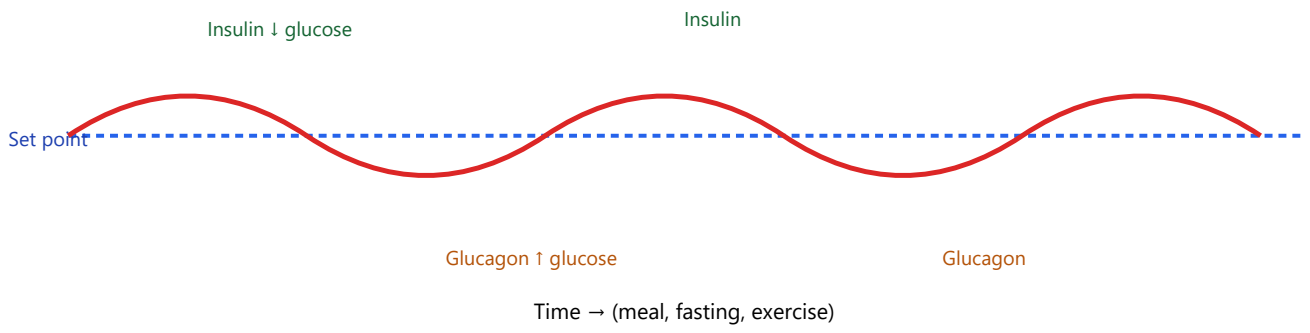

Fig 3: Negative feedback keeps blood glucose close to a set point by releasing insulin or glucagon.

Diabetes

Feature	Type 1	Type 2
Cause	Pancreas makes too little insulin	Cells resistant to insulin
Usual onset	Childhood	Adulthood; linked to obesity
Treatment	Insulin injections, carbohydrate control	Diet, exercise; drugs; sometimes insulin

4.5.8 Water & Nitrogen Balance (Separate Biology)

Water is gained from drinks, food, and respiration; lost in urine, sweat and exhaled air. Ions and urea must also be excreted.

Formation of Urea

Excess amino acids are broken down in the **liver** (*deamination*) producing toxic ammonia, which is converted to **urea** and transported in blood to the kidneys for excretion.

The Kidneys

Kidneys filter blood and reabsorb useful substances (glucose, selected ions, water as needed) while excreting urea and excess ions/water as **urine**.

ADH and Water Balance

Concentrated blood (dehydration) → osmoreceptors in hypothalamus → pituitary releases **ADH** → kidney collecting ducts become more permeable → more water reabsorbed → less, more concentrated urine.

Kidney Failure — Dialysis vs Transplant

Treatment	Pros	Cons
Dialysis	Available, no surgery	3× per week, strict diet, tied to machine, expensive long-term
Transplant	Long-term solution, normal life	Donor shortage; rejection risk; lifelong immunosuppressants

4.5.9 Reproduction

Menstrual Cycle

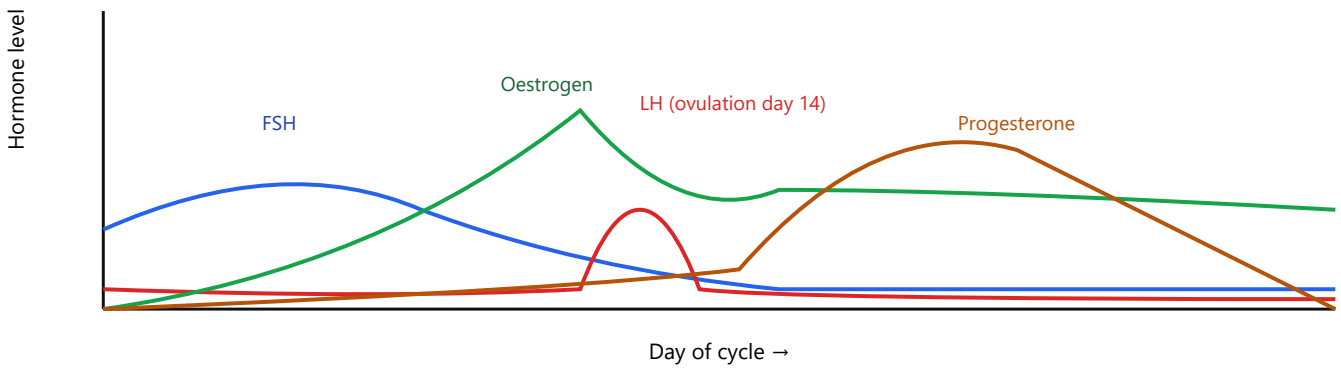

Fig 4: Hormone levels through a menstrual cycle. FSH matures a follicle; rising oestrogen triggers LH surge; LH causes ovulation. Progesterone maintains uterus lining.

Contraception

Method	How it works	Notes

Oral contraceptive pill	Oestrogen/progesterone inhibit FSH — no ovulation	Effective, but no STI protection
Injection/implant/patch	Slow release of progesterone	Longer lasting
Barrier (condom, diaphragm)	Prevents sperm reaching egg	Protects against STIs
IUD	Prevents implantation; hormonal or copper	Long-lasting
Spermicide	Kills sperm	Used with barrier
Abstinence, rhythm, surgical sterilisation	Various	Sterilisation is permanent

IVF (Separate Biology)

Used to treat infertility:

1. Mother given FSH and LH to stimulate egg maturation.
2. Eggs collected and fertilised with sperm in the lab.
3. Embryos develop for a few days.
4. One or two embryos implanted in uterus.

Success rates are low (~30%); drugs can cause side effects; ethical concerns about unused embryos; risk of multiple births.

4.5.10 Hormones in Plants (Separate Biology)

Auxin is a hormone produced in shoot/root tips that controls growth in response to light (phototropism) and gravity (gravitropism).

- **Shoot**: light from one side — auxin moves to the shaded side — cells there elongate more — shoot bends towards light (*positive phototropism*).
- **Root**: horizontal — auxin accumulates on lower side. In roots, high auxin *inhibits* elongation — root bends down (*positive gravitropism*).

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Commercial Uses of Plant Hormones

- **Auxins** — selective weedkillers, rooting powders, tissue culture.
- **Gibberellins** — end seed dormancy, promote flowering, make larger fruits.
- **Ethene** — ripens fruit during storage and transport.

> **Exam Tip:** For 6-mark questions on homeostasis, always name the stimulus, receptor, coordination centre, effector and response — and explain the negative feedback loop that restores the set point.

Summary — 4.5

- Homeostasis maintains optimum conditions (temp, glucose, water) using negative feedback.
- Nervous system: fast, targeted. Reflex arc = sensory → relay → motor.
- Endocrine system: slower, long-lasting. Pituitary = master gland.
- Blood glucose: insulin (lowers, stores as glycogen) vs glucagon (raises).
- Menstrual cycle: FSH → egg maturation; LH surge → ovulation; oestrogen & progesterone prepare lining.
- Contraception: hormonal (pill), barrier (condom), IUD, sterilisation.
- Plant tropisms controlled by auxin.

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